

DEVELOPMENT OF PROBABILISTIC MODELS FOR PREDICTING LONG RANGE CONTACTS IN PROTEIN STRUCTURES

CHAPTER 1 INTRODUCTION

1.1 Protein fold

Protein folds describe the unique three-dimensional arrangement of a protein's amino acid chain, determining its overall structure and function. Different fold types, such as all-alpha, all-beta, and mixed alpha/beta, have distinct patterns of residue interactions and spatial organization. Understanding these folds is essential for studying protein stability, function, and evolution. Protein fold classification helps group proteins with similar structures, providing insights into their biological roles. In structural biology, analyzing folds allows better prediction of residue contacts and distances, which is key for accurate protein modeling and drug design [1]–[3].

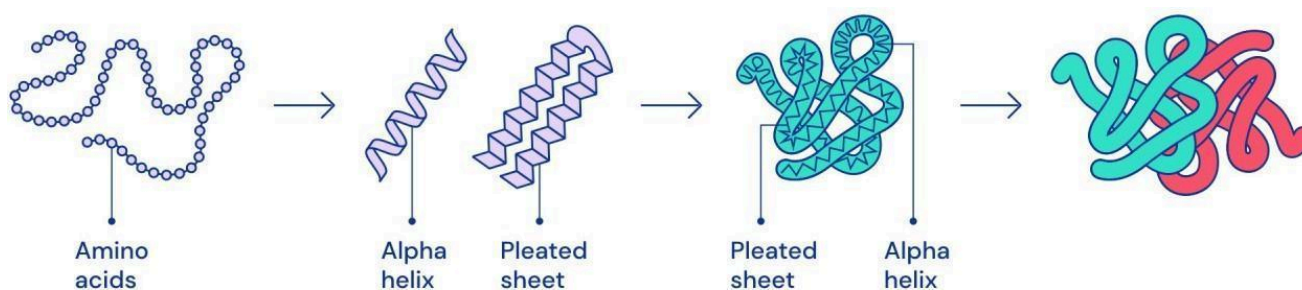


Fig 1. Folding of a protein

1.2 CATH and SCOP Databases

CATH and SCOP are widely used databases that classify protein structures based on their shapes and evolutionary relationships. SCOP organizes proteins hierarchically into classes, folds, superfamilies, and families through manual curation [2]. CATH classifies proteins by class, architecture, topology, and homologous superfamily using both automated and manual methods [3]. These classifications help researchers understand structural diversity and evolutionary links among proteins. Both databases are essential for studying protein folds and improving structure prediction [2], [3].

1.3 Neighboring atoms

Neighboring atoms are atoms located close to each other within a specific distance, often within 8 angstroms in protein structures. They play a key role in stabilizing the protein's 3D shape through interactions like hydrogen bonds and van der Waals forces [4]. Understanding which atoms are neighbors helps reveal how amino acids interact and fold, which is essential for predicting protein structure and function [1], [2]. Analyzing neighboring atoms provides insights into residue contacts and spatial organization. This information is vital for modeling protein folding and dynamics accurately. Identifying neighbors is a fundamental step in studying protein stability and interaction networks [2], [3], [4].

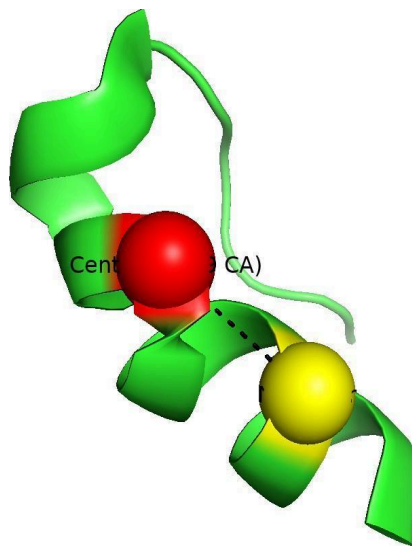


Fig 2. Neighboring atom within 8 angstroms (taken from pdb id: 1L2Y)

1.4 Residue–Residue Contacts

Residue–residue contacts refer to pairs of amino acid residues in a protein that are spatially close enough to interact, typically within a certain distance cutoff like 8 angstroms between their $C\alpha$ atoms [4], [5]. These contacts are critical for maintaining the protein's three-dimensional structure and are key determinants of protein folding and stability [1], [5]. Identifying residue contacts helps

understand how different parts of a protein come together to form functional sites and overall architecture. Accurate prediction of these contacts is essential for computational protein structure modeling and for designing drugs that target specific protein regions [5].

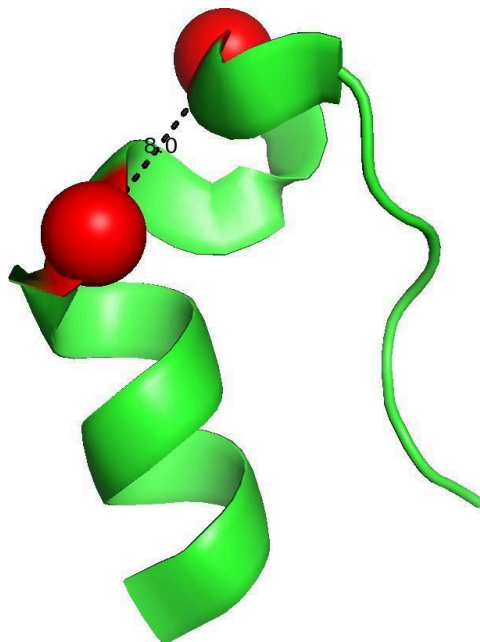


Fig 3. Residue-residue contact of res 9 and res 13 (taken from pdb id: 1L2Y)

1.5 Use of Protein–Protein Interactions in Statistical Analysis

Protein–protein interactions (PPIs) provide valuable data on how proteins interact at the residue level across different structural contexts. Analyzing these interactions helps identify recurring patterns in residue contacts, such as which amino acid pairs are more likely to interact and at what distances [6]. In this project, statistical data from PPIs can be used to enrich spatial frequency matrices, revealing the most common contact distances between residues in interacting proteins. This enhances the fitting of accurate probabilistic models by providing more diverse and realistic distance distributions, especially useful when modeling multimeric or complex protein structures [4], [6]. Including PPI data improves the robustness and generalizability of the distance prediction models [5], [6].

1.6 Types of Protein Folds

1.6.1 All-Alpha Fold

- Consists mainly of alpha helices packed together.
- **Example:** *Myoglobin* – a heme-binding protein composed almost entirely of alpha helices.

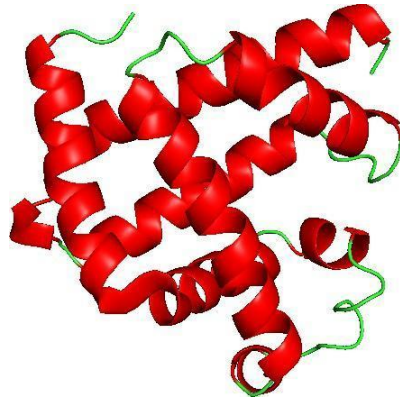


Fig 4. Structure of Myoglobin (taken from pdb id: 1A6M)

1.6.2 All-Beta Fold

- Composed primarily of beta sheets arranged in various patterns.
- **Example:** *Concanavalin A* – a lectin protein with a beta-sandwich structure.

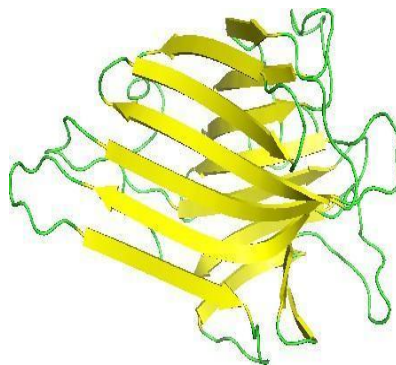


Fig 5. Structure of Concanavalin A (taken from pdb id: 3CNA)

1.6.3 Alpha/Beta Fold

- Alternating alpha helices and beta strands, often forming a beta-alpha-beta motif
- **Example:** *Triose phosphate isomerase (TIM barrel)* – contains a common alpha/beta barrel fold.

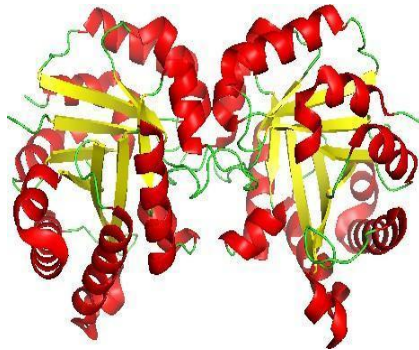


Fig 6. Structure of Triose phosphate isomerase (taken from pdb id: 1TIM)

1.6.4 Alpha + Beta Fold

- Contains separate regions of alpha helices and beta sheets that do not alternate.
- **Example:** *Ribonuclease A* – has distinct alpha and beta domains.

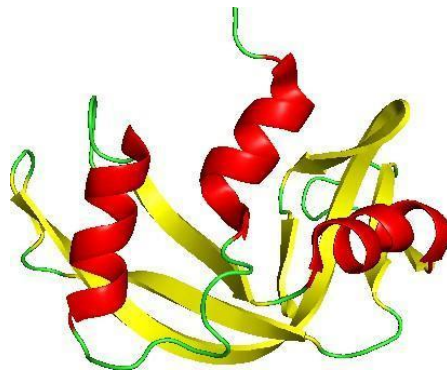


Fig 7. Structure of Ribonuclease A (taken from pdb id: 7RSA)

CHAPTER 2

OBJECTIVE

This project aims to develop a probabilistic model for predicting real-valued inter-residue distances in proteins that can:

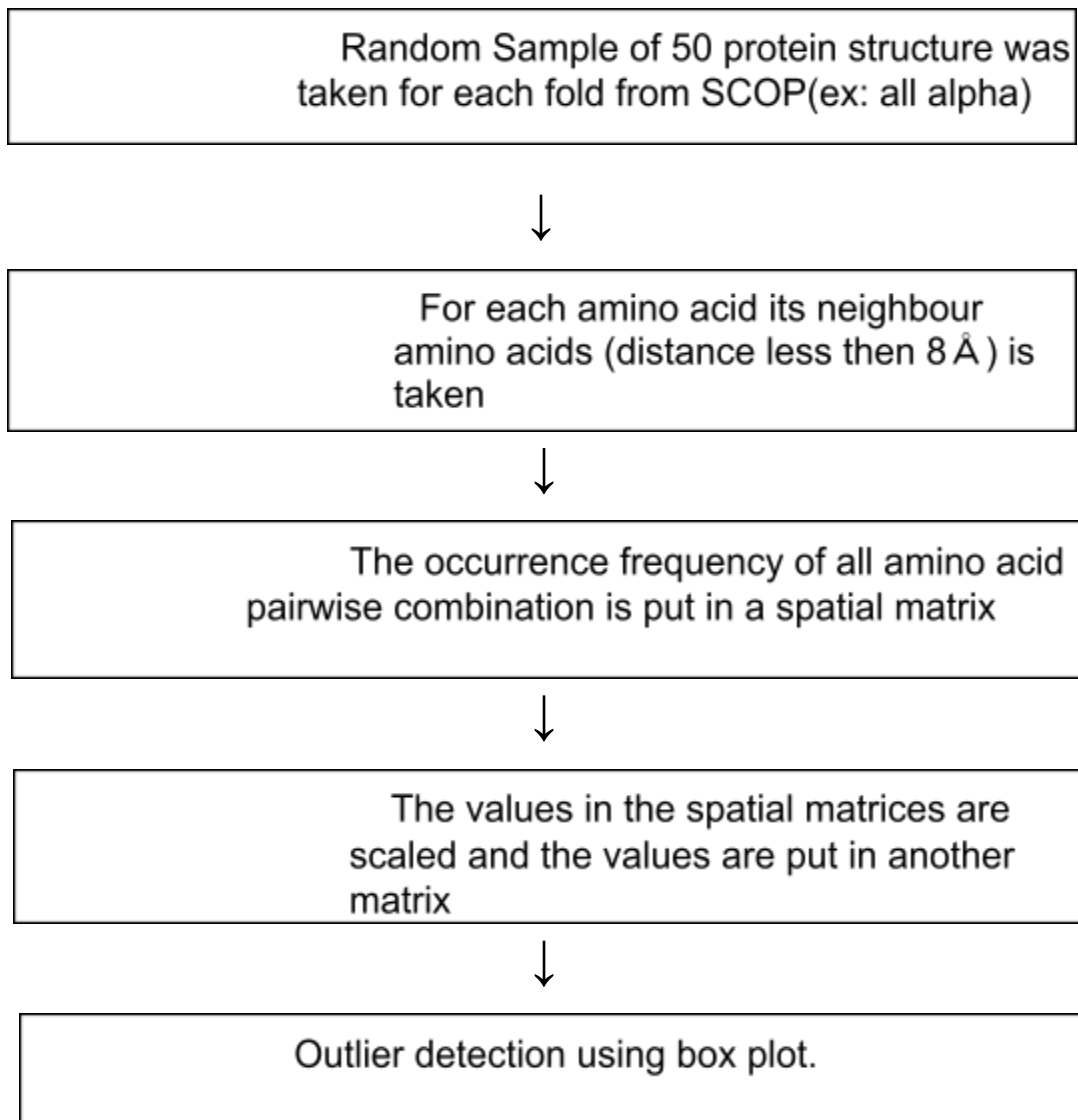
- Analyze protein structures to generate spatial frequency data of amino acid pairs for long-range contacts.
- Fit suitable statistical distributions to model these inter-residue distances accurately.
- Enable better prediction of residue interactions, supporting both monomeric and multimeric protein structure modeling.

This approach aims to enhance protein structure prediction precision and provide a flexible framework for applications in computational structural biology.

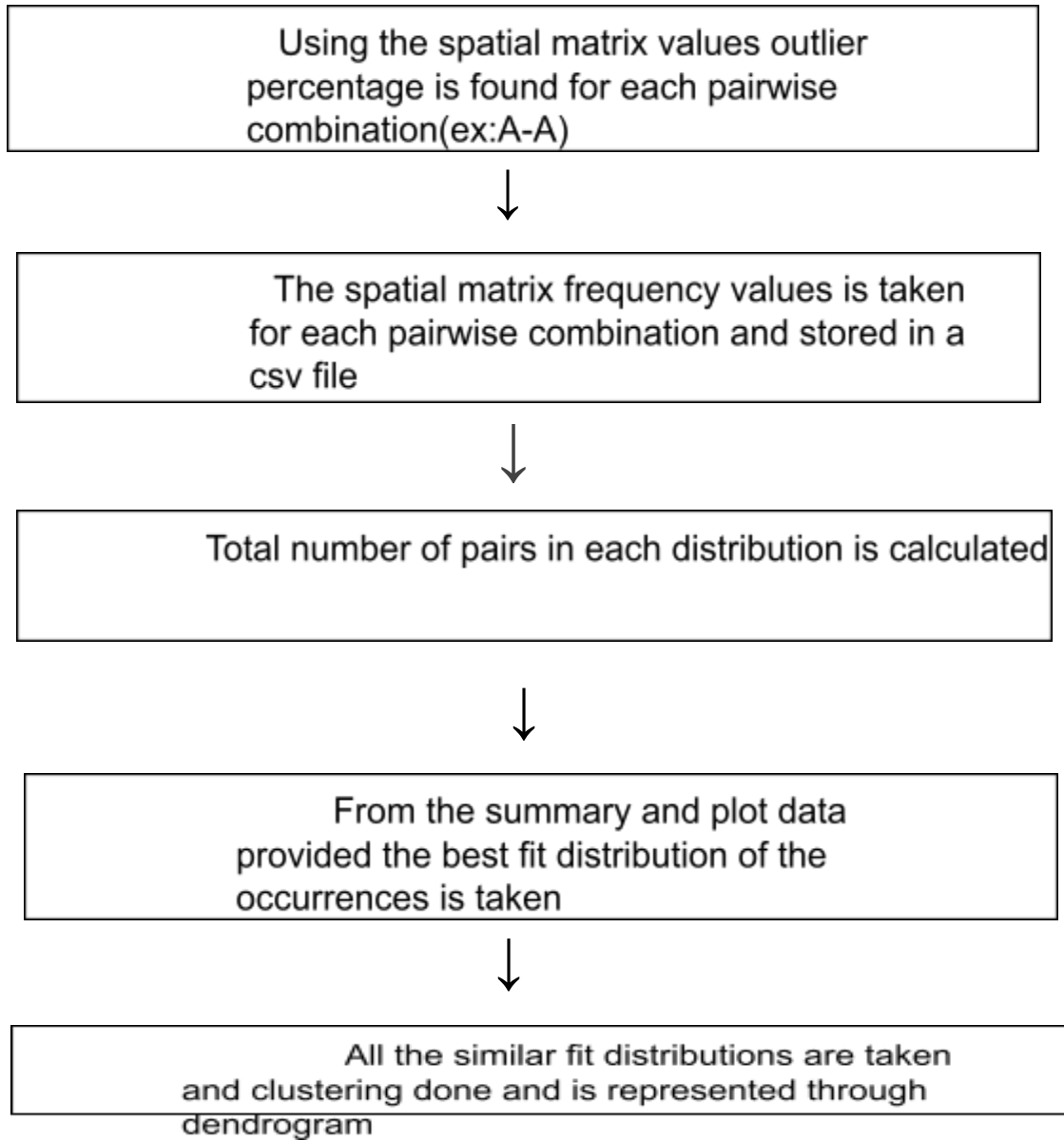
CHAPTER 3

METHODOLOGY

3.1 FLOW CHART



FLOW CHART CONTINUED



CHAPTER 4

RESULTS AND DISCUSSION

4.1 Collection of Random PDB IDs

A random set of 50 non-redundant, high-resolution protein structures was selected for each SCOP fold class (all-alpha, all-beta, alpha+beta, alpha/beta) to ensure quality and consistency in analysis.

1	PDB_ID	40	1DXX
2	2BCQ	41	1LNS
3	2FJ6	42	1RYK
4	1SET	43	1YWW
5	2IY5	44	2O6K
6	1HUZ	45	2JPN
7	2FMP	46	3L0O
8	4YUU	47	1XPU
9	4C47	48	1KCF
10	2XDJ	49	1V66
		50	1EN7
		51	5TCS

Fig 8. All alpha pdb ids

1	PDB_ID	40	1JQP
2	2GTL	41	5TTF
3	2JNG	42	3HNA
4	2JOY	43	2G46
5	2JUF	44	3IHX
6	3QKJ	45	3EP0
7	6KCP	46	3RAY
8	6IIT	47	6XAU
9	3EAE	48	1GTF
10	3LLR	49	3ZZL
		50	1YOX
		51	1WAP

Fig 9. All beta pdb ids

1	PDB_ID	40	4LRS
2	1HL9	41	4JN6
3	4PSR	42	7JW9
4	1U2Y	43	1LUC
5	1R46	44	3B4Y
6	1DH2	45	5W48
7	1ESW	46	1NQK
8	5H2T	47	3SDO
9	3WY1	48	5LXE
10	4MAZ	49	1RHC
		50	6AK1
		51	3RAO

Fig 10. Alpha/beta pdb ids

1	PDB_ID	40	1J2G
2	2CYY	41	2IUT
3	2CFX	42	2IUS
4	2CG4	43	1WQG
5	1I1G	44	3KSF
6	1S7I	45	1TIG
7	2FIU	46	2BVC
8	3LO3	47	1RQ8
9	5KAK	48	5UCG
10	2AP6	49	2D9I
		50	1F5M
		51	3SEB

Fig 11. Alpha+beta pdb ids

4.2 Generation of Spatial and Normalized Matrices for each structure

- For each amino acid pair its occurrence within 8 angstroms was counted for each structure and the result is stored as in a spatial matrix.
- The same process is repeated for every fold type.

	A	B	C	D	E	F	G	H
1		A	C	D	E	F	G	H
2	A	24	2	7	17	15	5	2
3	C	2	0	0	1	2	1	0
4	D	7	0	0	5	4	1	0
5	E	17	1	5	6	5	4	2
6	F	15	2	4	5	4	0	2
7	G	5	1	1	4	0	0	0
8	H	2	0	0	2	2	0	0

Fig 12. Spatial Matrix for the pdb id

- The values are normalized and are stored in a normalized matrix.
- The same process is repeated for each structure in every fold.

	A	B	C	D	E	F	G	H
1		A	C	D	E	F	G	H
2	A	0.02439	0.002033	0.007114	0.017276	0.015244	0.005081	0.002033
3	C	0.002033	0	0	0.001016	0.002033	0.001016	0
4	D	0.007114	0	0	0.005081	0.004065	0.001016	0
5	E	0.017276	0.001016	0.005081	0.006098	0.005081	0.004065	0.002033
6	F	0.015244	0.002033	0.004065	0.005081	0.004065	0	0.002033
7	G	0.005081	0.001016	0.001016	0.004065	0	0	0
8	H	0.002033	0	0	0.002033	0.002033	0	0

Fig 13. Normalized Matrix for the pdb id

4.3 Outlier Detection using Boxplots

- Using the normalized matrix values for each pair of amino acids, a boxplot is generated to identify the outliers present.
- The same process is repeated for all the 400 pairwise amino acid combinations for all the folds.

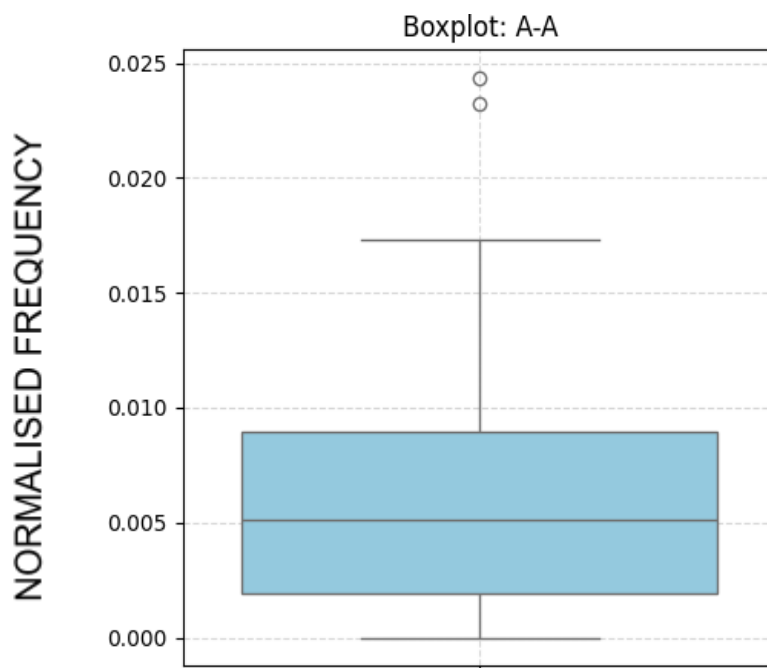


Fig 14. Boxplot for AA amino acid pair(all alpha)

4.4 Outlier Statistics for each Amino Acid pair

- Using z-score the amino acid pair which contains outliers is alone taken along with its structure id, and the count of outliers present.

	A	B	C	D
1	AminoAcid	PDB_ID	Count	Z_Score
2	A-A	1IWP	106	3.228
3	A-A	1EGV	94	2.78
4	A-A	6L35	92	2.706
5	A-A	6K61	78	2.184
6	A-A	7S3D	76	2.109
7	A-C	1IWP	9	3.52

Fig 15. Outlier count and z-score of outliers containing amino acid pairs

- Using Mean, standard deviation the outlier percentage is calculated.

	A	B	C	D	E	F	G
1	AminoAcid	Mean	StdDev	Outlier_Count	Total_Structure	Outlier_Percentage	
2	Q-T	4.74	5.527	6	50	12	
3	A-A	19.44	26.818	5	50	10	
4	D-W	1.66	2.094	5	50	10	
5	D-G	6.12	7.191	5	50	10	
6	C-I	1.14	1.866	5	50	10	
7	C-Q	0.94	1.462	5	50	10	
8	W-Y	1.7	2.46	5	50	10	

Fig 16. Outlier percentage calculation for outliers containing pairs in each structure

4.5 Overall Outlier summary

- The overall outlier summary containing the outlier percentage is ordered in descending order and stored in a csv file format.
- This file is for one fold type the same is repeated for all fold types.

	A	B	C	D	E
1	AminoAcid	Total_Occ	Outlier_Cc	Outlier_Percentage	
2	H-H	15	5	33.33	
3	C-E	13	4	30.77	
4	C-Q	21	5	23.81	
5	H-W	17	4	23.53	
6	C-I	22	5	22.73	
7	C-P	14	3	21.43	
8	I-W	24	5	20.83	

Fig 17. Overall outlier summary for each fold

4.6 Pairwise values across structures

- The amino acid pairwise frequency counts across each structure for each pair is put as a matrix in a separate csv file.

	A	B	C	D	E	F	G	H
1	AminoAcid	Struct_1	Struct_2	Struct_3	Struct_4	Struct_5	Struct_6	Struct_7
2	A-A	24	4	94	2	6	106	10
3	A-R	7	6	40	0	13	41	5
4	A-N	5	12	27	7	7	22	12
5	A-D	7	4	31	3	7	29	12
6	A-C	2	0	8	2	1	9	2
7	A-Q	5	8	28	6	8	33	7
8	A-E	17	3	39	6	6	55	6
9	A-G	5	8	56	4	7	45	9
10	A-H	2	3	8	2	1	2	0

Fig 18. Pairwise frequency across 50 structures in a fold

4.7 Generation of fit summary

- For each pair fit distribution is done using R.
- Fit summary is generated for each pair in a fold.

```

=== Summary for A-A ===
Fitting of the distribution ' unif ' by maximum likelihood
Parameters :
      estimate
min          2
max         106
Loglikelihood: -181.1312   AIC: 366.2625   BIC: 369.5896
Fitting of the distribution ' norm ' by maximum likelihood
Parameters :
      estimate Std. Error
mean 24.92308   4.487602
sd   28.02507   3.173214
Loglikelihood: -185.3295   AIC: 374.659   BIC: 377.9861
Correlation matrix:
      mean sd
mean    1  0
sd      0  1

Fitting of the distribution ' gamma ' by maximum likelihood
Parameters :
      estimate Std. Error
shape 0.87240180 0.171699464
rate  0.03501053 0.009127822
Loglikelihood: -164.1675   AIC: 332.3349   BIC: 335.662
Correlation matrix:
      shape    rate
shape 1.0000000 0.7538631
rate  0.7538631 1.0000000

```

Fig 19. Fit summary for AA amino acid pair

- The statistical summary for each pair is compared with the AIC values from each distribution.
- The distribution with low AIC value is taken as the best fit distribution.

	A	B	C	D	E	F	G	H
1	AminoAcid	Normal_A	Exponential	Gamma_A	Best_Fit	Gamma_S	Gamma_R	Exp_Rate
2	A-A	474.8013	398.7333	332.3349	Gamma	0.872689	0.035014	0.05144
3	A-R	376.8256	317.4085	292.7593	Gamma	1.043162	0.106478	0.116009
4	A-N	346.1061	305.3398	279.219	Gamma	1.417934	0.163348	0.13089
5	A-D	354.3544	314.1063	295.5164	Gamma	1.433581	0.158163	0.119904

Fig 20. Statistical summary with AIC values for each pair

4.8 Generation of Cullen-Frey plot

- For each pair Cullen- Frey plot is generated to aid the selection of best distribution.

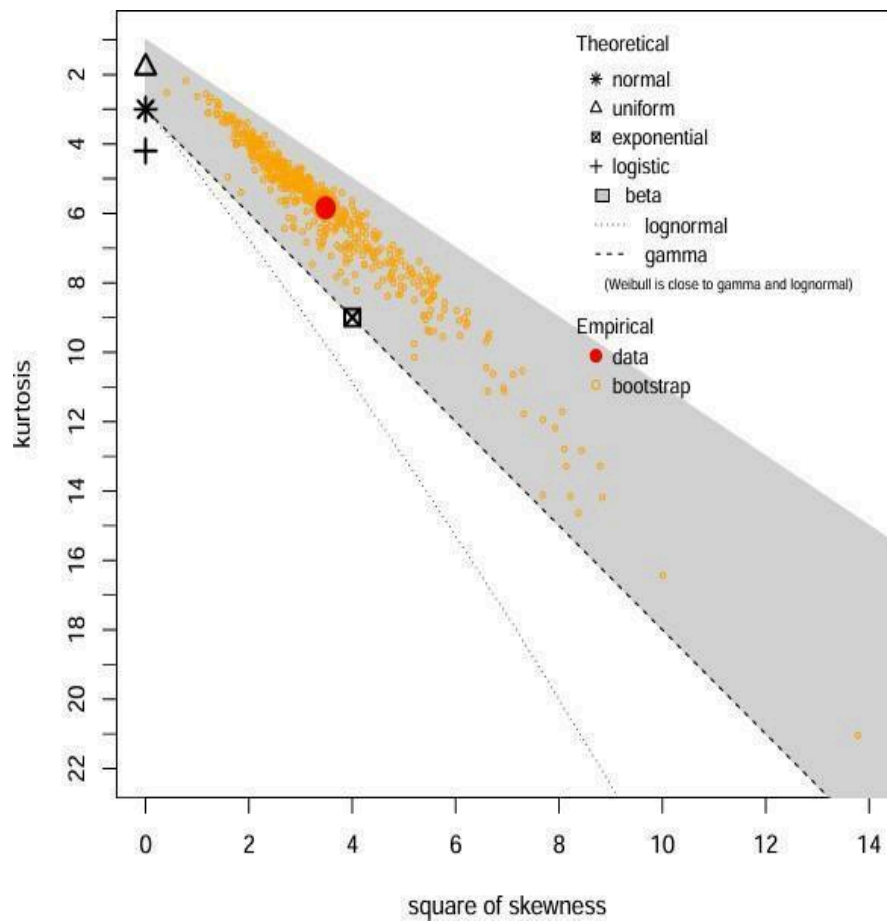
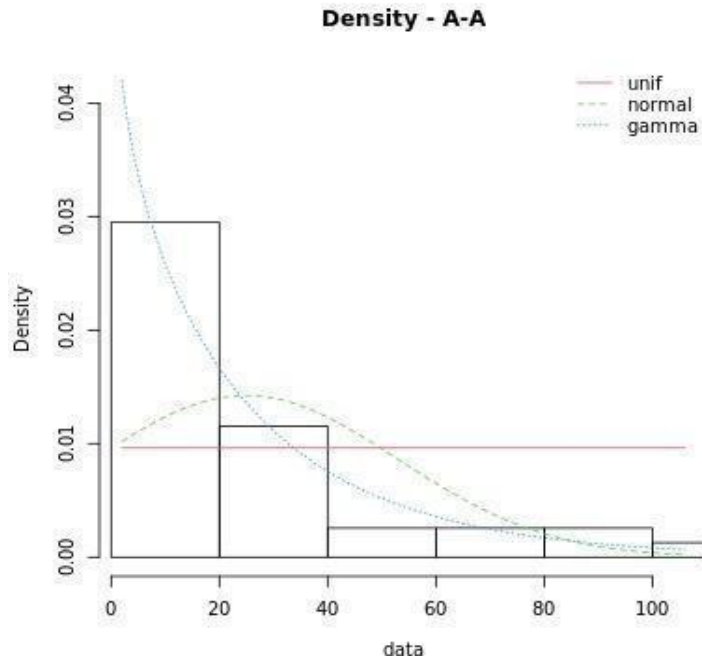


Fig 21. Cullen - Frey plot for AA amino acid pair

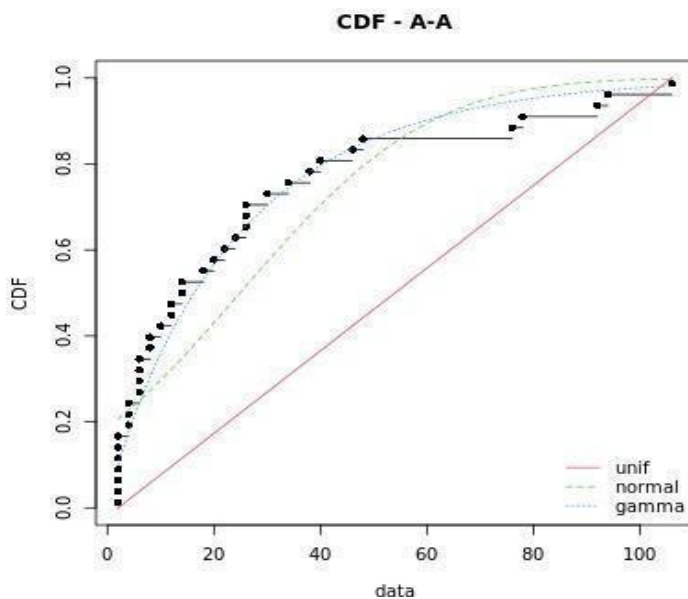
4.9 Generation of Density plots

- For each pair, plots like Density, CDF, QQ and PP plots are generated inorder to validate the best fit for an amino acid pair.



Density plot: Histogram shows the actual distribution of A-A data.

Fig 22. Density plot for A-A pair



CDF plot: Black dots and steps - empirical CDF data of the A-A data.

Fig 23. CDF plot for A-A pair

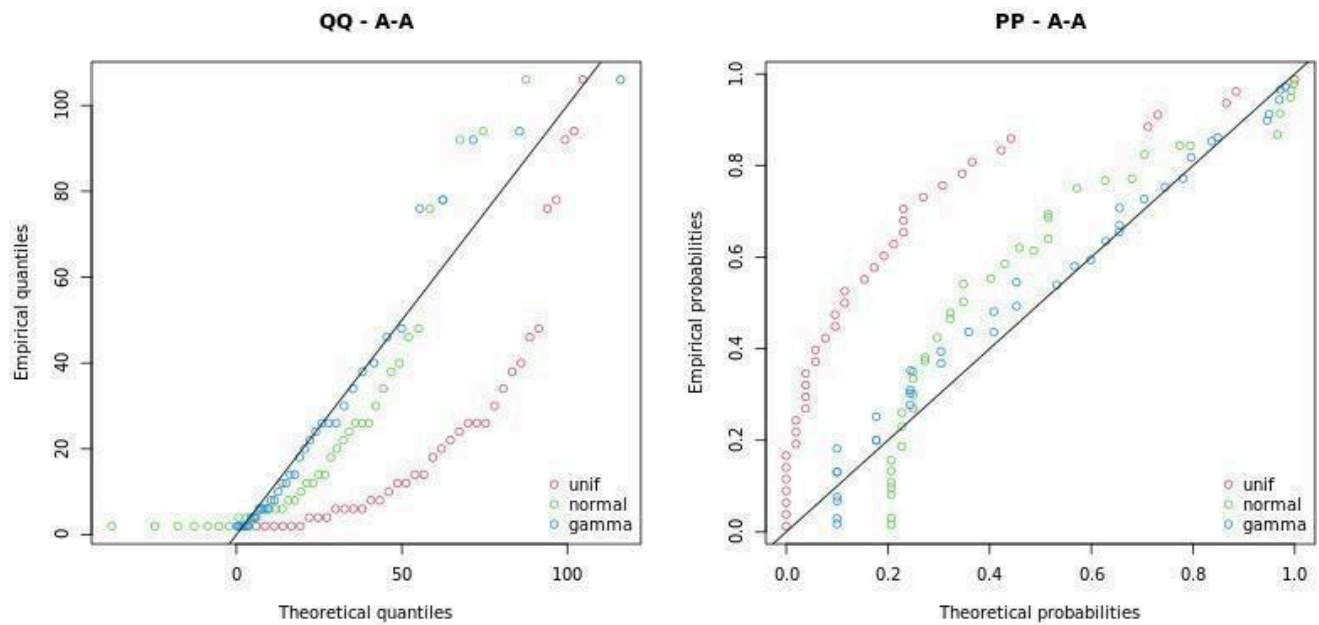


Fig 24. QQ and PP plots for A-A pair

- QQ plot: Shows that the gamma distribution closely matches the quantiles of the A-A data indicating a good fit.
- PP plot: Shows that the gamma distribution also best matches the cumulative probabilities of the data confirming its suitability.

4.10 Creation of dendrogram

- Using clustering techniques the gamma distribution of a fold is taken and clustered based on the gamma parameters and is visualised using dendrogram.
- The same is repeated for whatever distribution fits the pairs.
- In this case mostly Gamma, Exponential and normal distribution is fitted best in most cases.

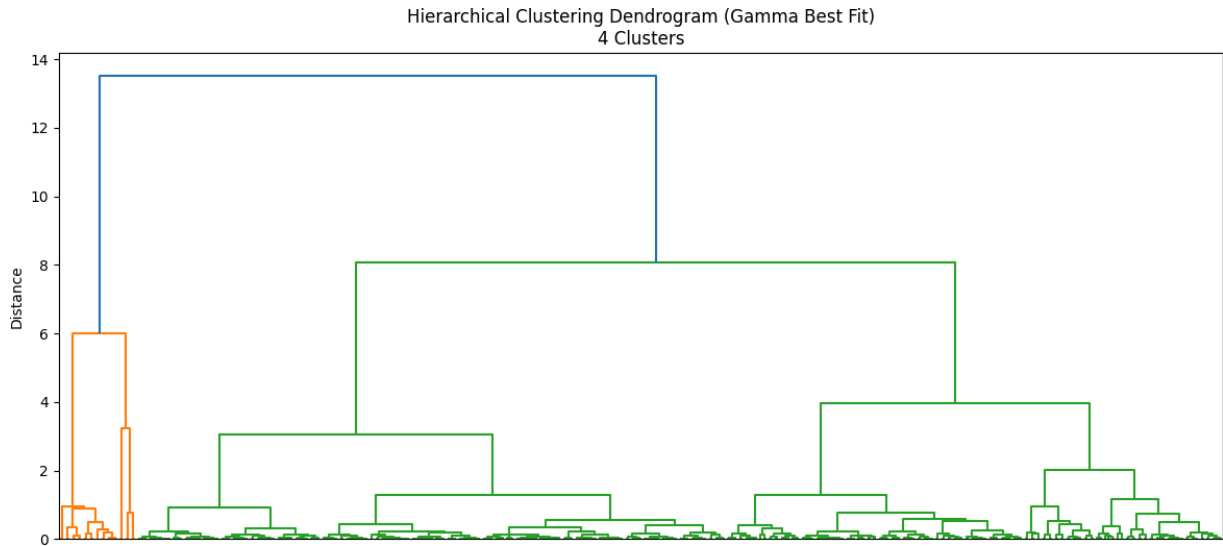


Fig 25. Dendrogram for gamma distribution in all alpha fold

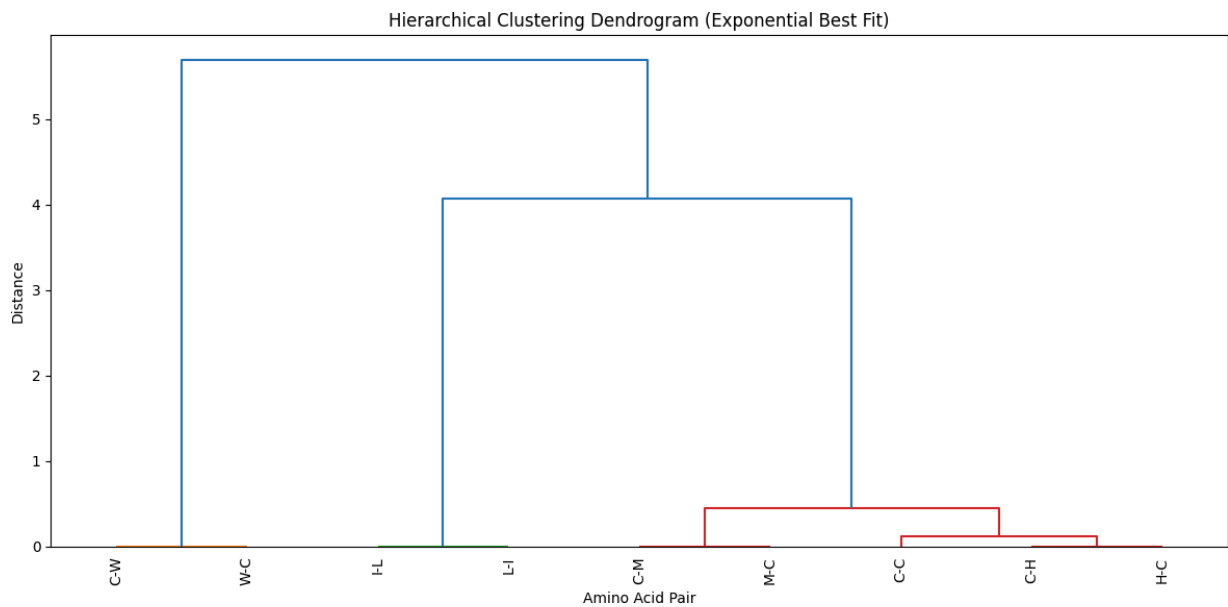


Fig 26. Dendrogram for exponential distribution in all alpha fold

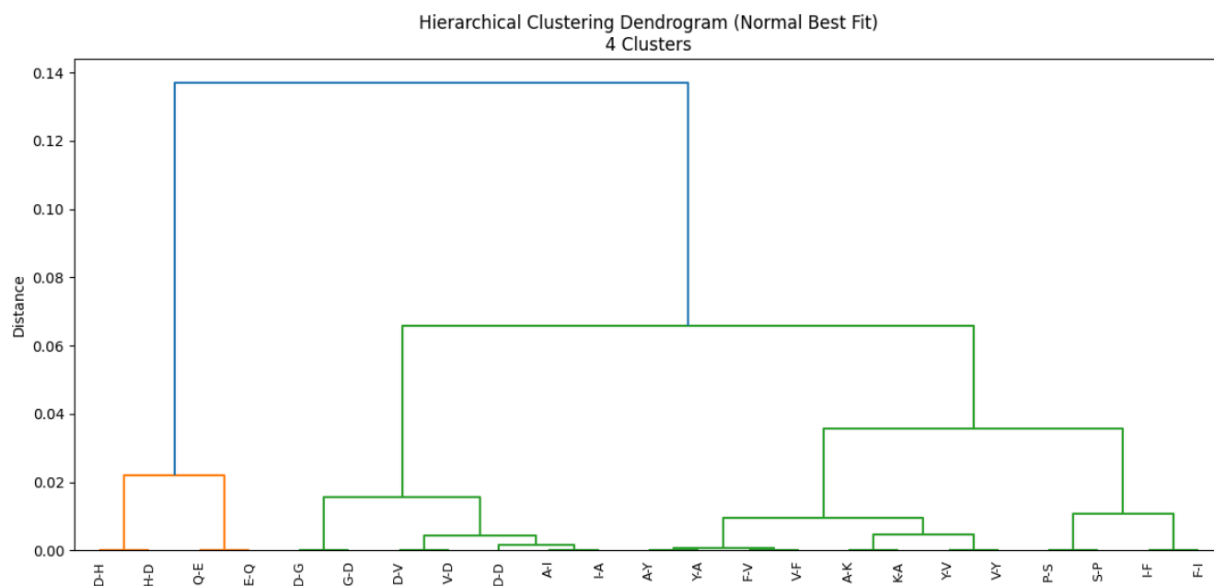


Fig 27. Dendrogram for normal distribution in alpha/beta fold

4.11 Total Occurrence of each fit distribution across all folds

Total Occurrence of each best distribution	
GAMMA	1539
EXPONENTIAL	38
NORMAL	23

Tab 1. Total Occurrence of each best distribution

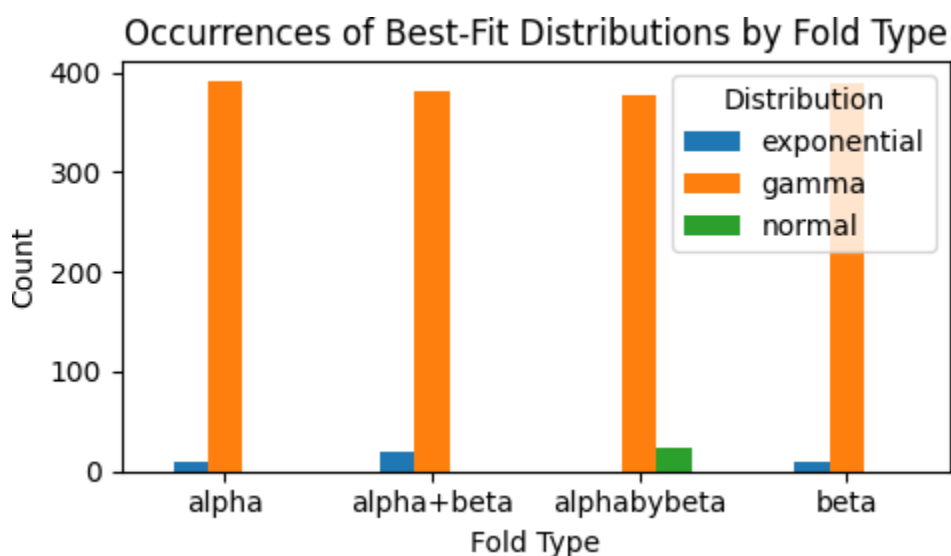


Fig 28. Bar graph for the occurrences of Best Fit Distribution by fold type

4.12 Distribution of each fit distribution across all folds

- The distribution of each best fit distribution across all folds is analysed using a pie chart.

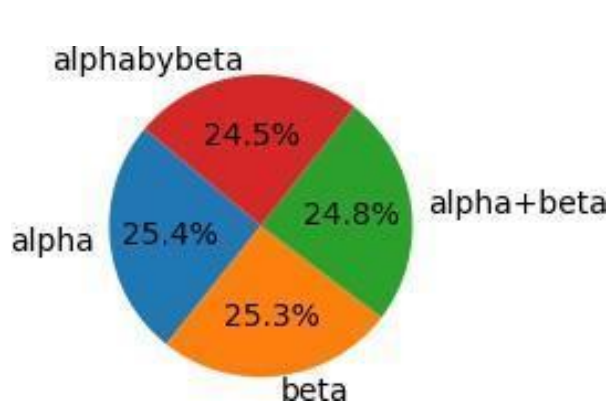


Fig 29. Pie chart showing the gamma distribution

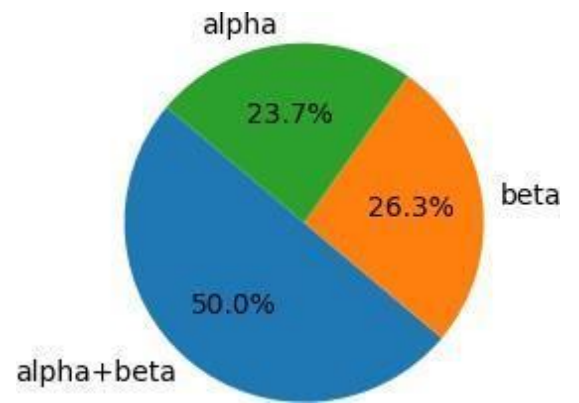


Fig 30. Pie chart showing the exponential distribution



Fig 31. Pie chart showing the normal distribution

CHAPTER 5

CONCLUSION

In this project, we developed a probabilistic approach to improve the prediction of long-range residue contacts in protein structures. Instead of using traditional binary methods, we analyzed the actual distances between amino acid pairs within 8Å across different protein fold types—All-Alpha, All-Beta, Alpha+Beta, and Alpha/Beta.

We created spatial and normalized matrices of pairwise amino acid distances and applied statistical methods to find the best-fitting distributions. Among the distributions tested (Gamma, Exponential, and Normal), the **Gamma distribution** was found to be the most common best fit, with **1539 occurrences**, compared to only 38 for Exponential and 23 for Normal.

This shows that gamma distribution is well-suited to model the natural variation in residue distances found in protein structures. It also performed consistently across all fold types, making it a reliable model for predicting residue interactions.

The results were supported by various visualizations such as density plots, QQ/PP plots, bar graphs, and pie charts. These confirmed the strong performance of the gamma distribution in capturing the patterns in our data.

Overall, our work shows that using probabilistic models, especially the gamma distribution can make distance-based predictions more accurate and meaningful. This approach can be useful in future research for protein structure prediction, including deep learning applications and drug design.